Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python microservices**, **distributed systems**, and reliable **Azure/AWS identity-security** products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Achieved 99%+ production reliability across 5K+ monthly jobs through **data validation**, reconciliation checks, error handling, and CI/CD releases, connecting FastAPI services, relational databases, and cloud storage with incremental processing.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preventing cascading failures while maintaining recoverable, observable background processing under concurrent workloads.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** serving 150+ instructors and 10K+ daily grading requests, integrating secure ingestion, REST APIs, AWS RDS, and CI/CD-backed application workflows.
- Enabled fault-tolerant processing of 1K+ files per batch using **Python multithreading**, validating PDF/DOCX/ZIP inputs and isolating failures while preserving responsive instructor workflows during concurrent assessment processing and model execution.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads, troubleshooting cross-platform integrations while reducing technical debt by 20% for maintainable low-bandwidth product delivery.
- Increased pipeline efficiency by 68% through **AWS Glue and S3**; automated executable builds and 20+ unit/integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Supported synchronization of 2M+ records through a **microservice-based CRM platform**, defining services for authentication, role-based access, customers, billing, resource allocation, and notifications with React, Node.js, WebSockets, and Azure SQL.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, Azure monitoring, automated tests, and Docker/Jenkins deployments across eight modular product service domains.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a **distributed authentication architecture** spanning user, IoT-device, and role-based access scenarios, evaluating smart-contract verification, immutable identity records, and self-sovereign credential models for reducing reliance on centralized identity stores.
- Published peer-reviewed security research after analyzing **identity attack protections** against replay, phishing, man-in-the-middle, and credential theft, documenting access-control trade-offs and presenting technically defensible decentralized authentication designs at IEEE conferences.

Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Bash
Backend & Systems: Microservices, FastAPI, Flask, REST APIs, Distributed Systems, System Design, Multithreading, Fault Isolation, OOP
Identity & Security: IAM, User Authentication, RBAC, JWT, Data Validation, Audit Logging
Cloud & Data: Microsoft Azure, Azure SQL, AWS S3, AWS Glue, AWS RDS, GCP, PostgreSQL, SQL Server, MongoDB
Engineering: Unit Testing, Integration Testing, PyTest, Mockito, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes

Projects

Identity Recovery Orchestrator - Design Concept

- Proposed a **Python recovery microservice** for dependency-aware restoration of users, groups, roles, and service identities from immutable snapshots, outlining idempotent workflows, least-privilege authorization, audit logs, retries, and rollback validation.

Identity Drift Analyzer - Design Concept

- Proposed an **Azure identity analysis** pipeline that compares snapshots, detects privilege and relationship drift, prioritizes risky changes, and exposes review APIs with scalable storage, reconciliation tests, and customer-environment diagnostics.

Blockchain Authentication Research | Paper

- Evaluated **identity and access management** architectures for users, IoT devices, and role-based authorization, analyzing immutable records, smart-contract verification, self-sovereign identity, and secure credential sharing in a peer-reviewed IEEE publication.